RO2 - Lecture 3
3D-Accelerometer (Part 1)

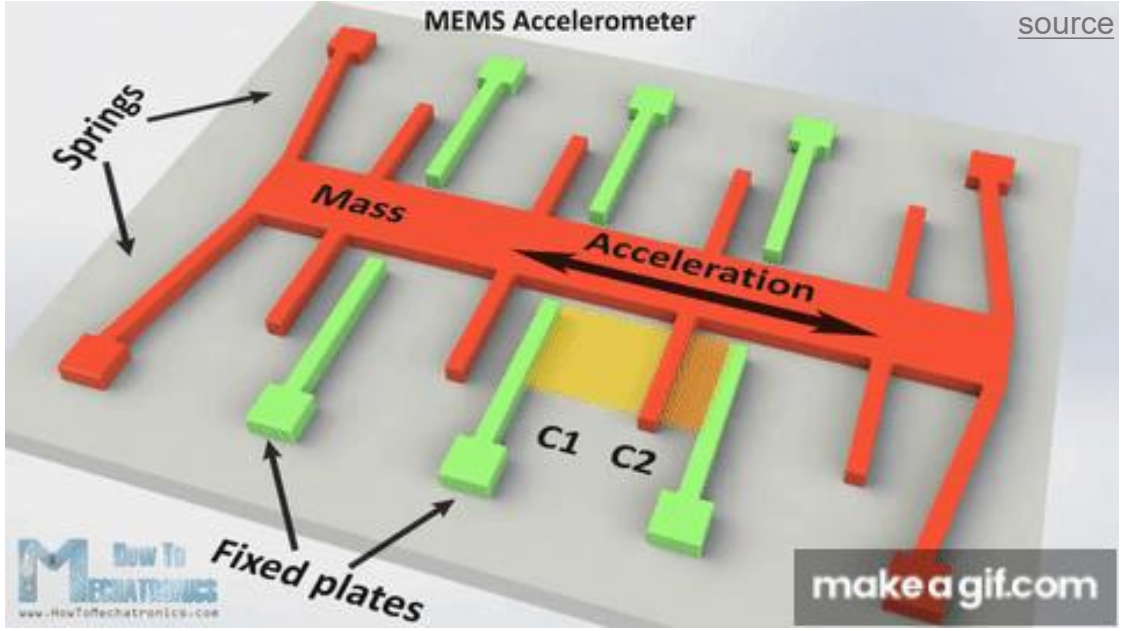
we focus on students

Fachhochschule Dortmund
University of Applied Sciences and Arts

3D Accelerometer

3D acceleration data – MEMS sensors

- MEMS (Micro-Electro-Mechanical Systems) accelerometer consists of a small, movable mass (proof mass) suspended by springs
- electrodes placed on the movable mass and fixed parts of the sensor create a capacitor
- As the accelerometer experiences acceleration, the movable mass shifts relative to the fixed electrodes
- the change in capacitance is proportional to the acceleration experienced by the device
- It is measured by an electronic circuit, which converts it into a corresponding voltage signal that can be used to quantify acceleration



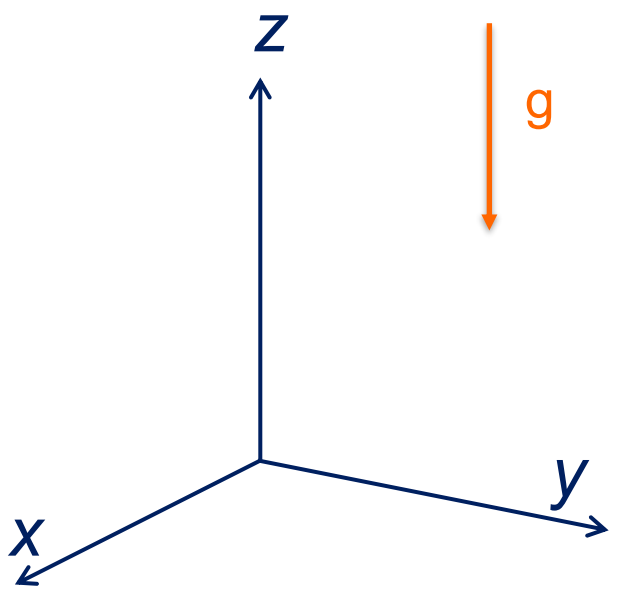
source

3D Accelerometer

3D acceleration data

Measure acceleration in m/s^2 ($\mathrm{a} = \Delta v/\Delta t$, unit of Δv = m/s, unit of Δt is s)

- If sensor is not moved (***stationary*** condition), it measures the gravity vector with magnitude g = 9.81m/s^2 :

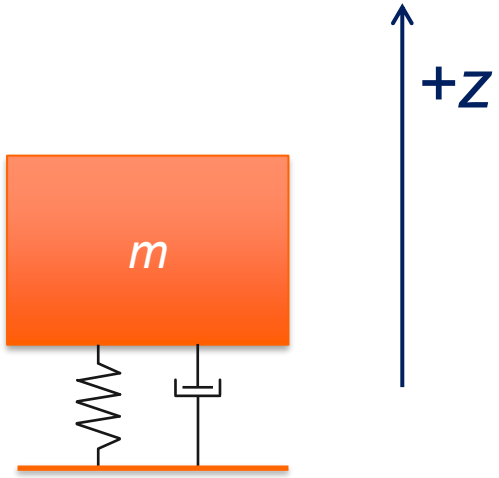


$$\boldsymbol{a}_{\mathrm{m}} = [0 \quad 0 \quad g]$$

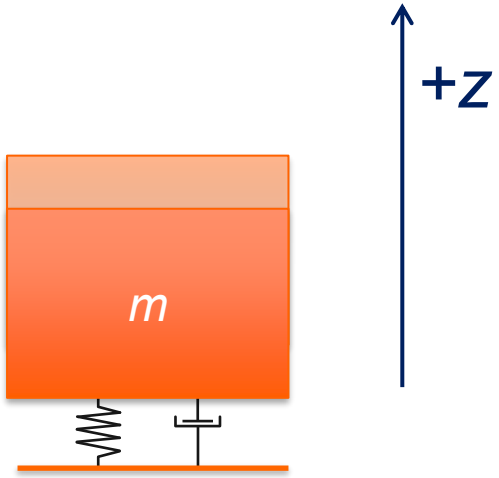
Really +g??

3D Accelerometer

Simplified model of MEMS accelerometer [1]



Simplified model of MEMS accelerometer [1]



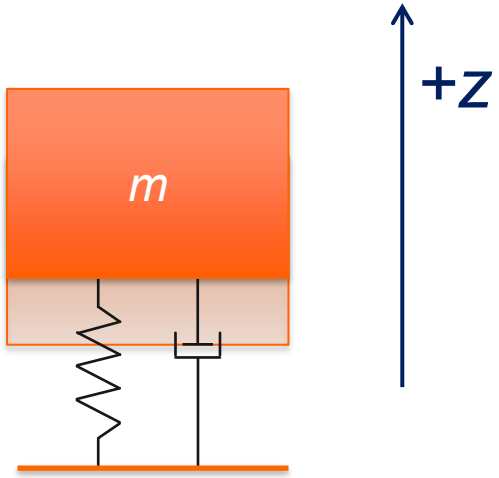
Acceleration upwards (along +z-axis direction):

spring-damper system compressed (due to mass inertia)

Compression creates positive values proportionial to applied acceleration (opposite direction of acceleration)

positive acceleration measurement value regarding z-axis

Simplified model of MEMS accelerometer [1]

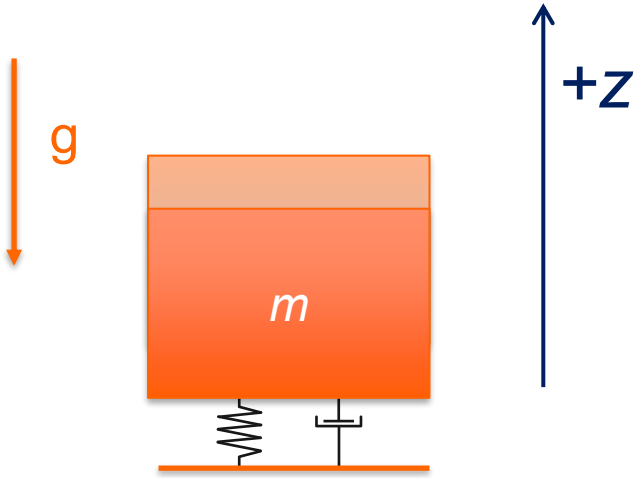


Acceleration downwards (along -z-axis direction):

spring-damper system elongated (mass pulls upwards)

negative acceleration measurement value regarding z-axis

Simplified model of MEMS accelerometer [1]



Stationary condition (z-axis parallel to gravity):

spring-damper system is compressed due to gravity

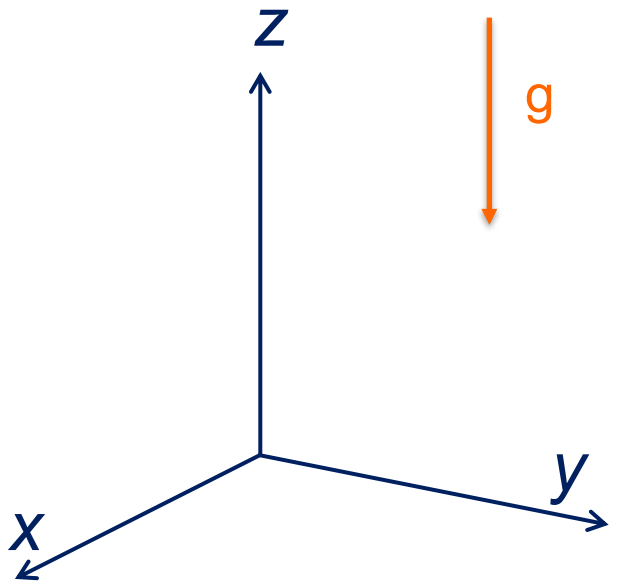
This matches the case of upward acceleration

positive acceleration measurement value regarding z-axis

3D Accelerometer

3D acceleration data

- Measure acceleration in m/s^2 ($\mathrm{a} = \Delta v/\Delta t$, unit of Δv = m/s, unit of Δt is s)
- If sensor is not moved (***stationary*** condition), it measures the gravity vector with magnitude g = 9.81m/s^2 :

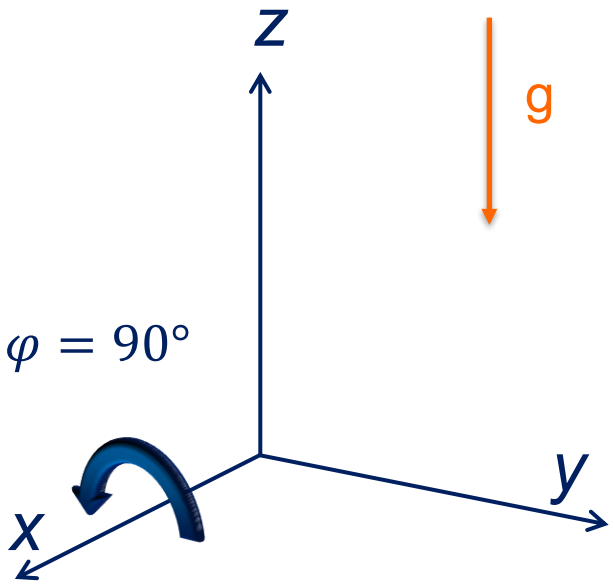


$$\boldsymbol{a}_{\mathrm{m}} = [0 \quad 0 \quad g]$$

✓

3D acceleration data

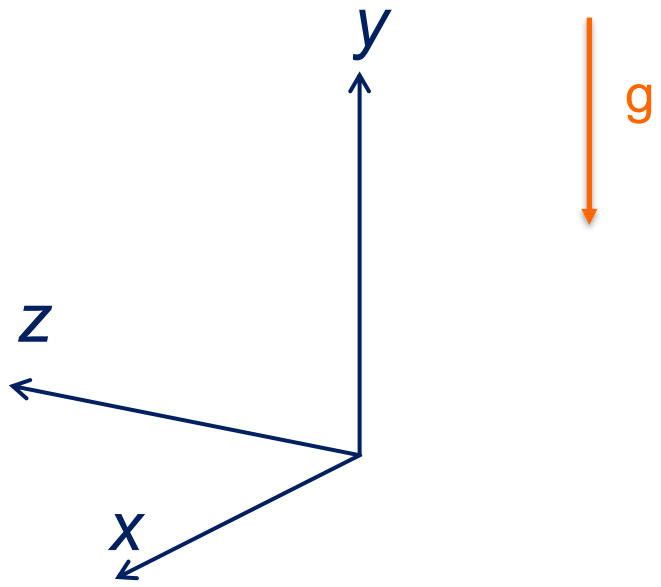
- How does the reading change when the sensor is rotated by 90° about the *x*-axis?



$$\boldsymbol{a}_{\mathrm{m}} = [0 \quad 0 \quad g]$$

3D acceleration data

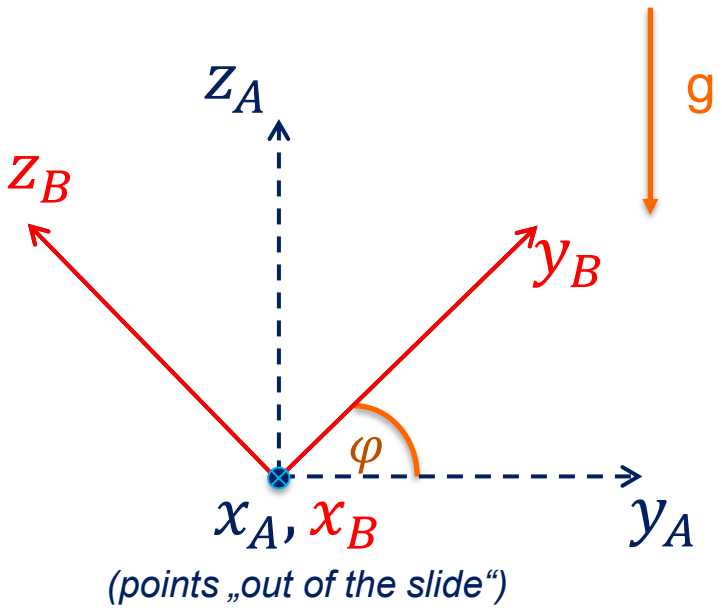
- How does the reading change when the sensor is rotated by 90° about the *x*-axis?



$$\boldsymbol{a}_{\mathrm{m}} = [0 \quad g \quad 0]$$

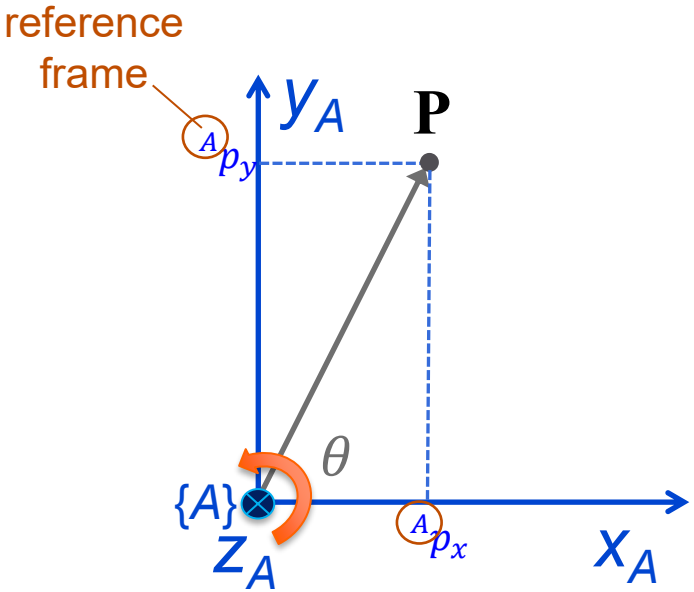
3D acceleration data

- How does the reading change when the sensor is rotated by an **arbitrary angle** φ about the *x*-axis?



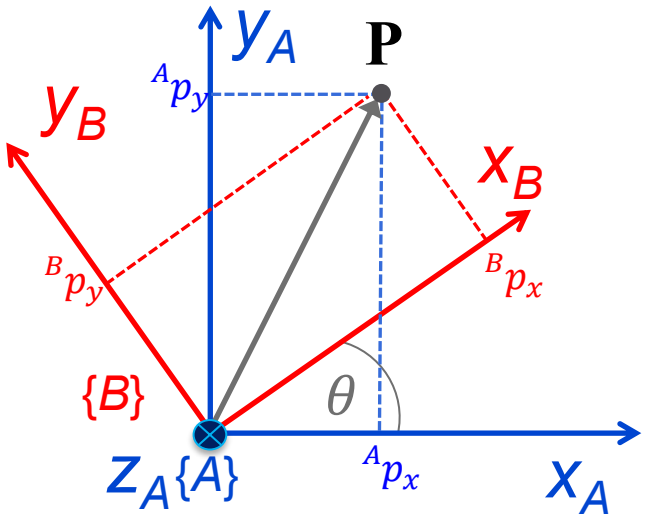
$${}^{B}\boldsymbol{a}_{\mathrm{m}} = [?\quad ?\quad ?]$$

Describing a vector before and after rotation



$$ {}^{A}\boldsymbol{p} = \begin{bmatrix} {}^{A}p_x \\ {}^{A}p_y \\ {}^{A}p_z \end{bmatrix} $$

Describing a vector before and after rotation



$$^A\boldsymbol{p} = \begin{bmatrix} ^A p_x \\ ^A p_y \\ ^A p_z \end{bmatrix} \qquad ^B\boldsymbol{p} = \begin{bmatrix} ^B p_x \\ ^B p_y \\ ^B p_z \end{bmatrix}$$

$$^A\boldsymbol{p} = {}^A\boldsymbol{R}_B \; {}^B\boldsymbol{p}$$

$$^B\boldsymbol{p} = {}^A\boldsymbol{R}_B^{-1} \; {}^A\boldsymbol{p} = {}^B\boldsymbol{R}_A \; {}^A\boldsymbol{p}$$

in general for a vector $\boldsymbol{v}$:

$$\boxed{^A\boldsymbol{v} = {}^A\boldsymbol{R}_B \; {}^B\boldsymbol{v}}$$

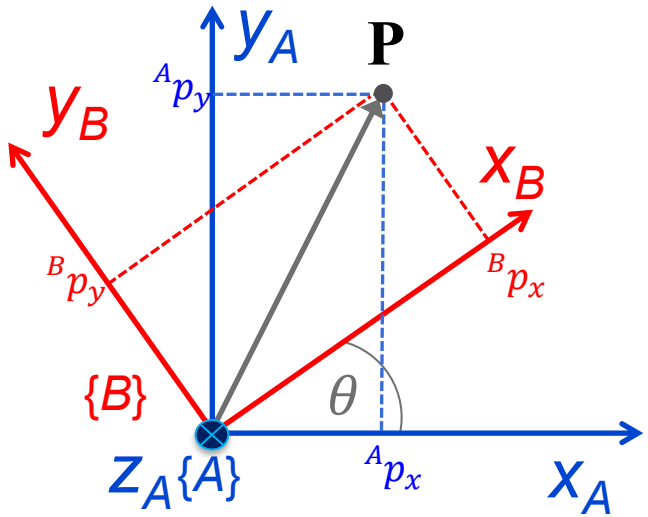
$$\boxed{^B\boldsymbol{v} = {}^A\boldsymbol{R}_B^{-1} \; {}^A\boldsymbol{v} = {}^B\boldsymbol{R}_A \; {}^A\boldsymbol{v}}$$

Describing a vector before and after rotation

From last lecture:

$$\boldsymbol{q}^{-1} = \boldsymbol{q}^{*} = [q_0 \quad -q_1 \quad -q_2 \quad -q_3]$$

$$\left({}^{B}\boldsymbol{q}_{A}\right)^{-1} = {}^{A}\boldsymbol{q}_{B}$$



$${}^{A}\boldsymbol{p} = \begin{bmatrix} {}^{A}p_x \\ {}^{A}p_y \\ {}^{A}p_z \end{bmatrix} \qquad {}^{B}\boldsymbol{p} = \begin{bmatrix} {}^{B}p_x \\ {}^{B}p_y \\ {}^{B}p_z \end{bmatrix}$$

With quaternions:

1. make quaternion from vector $\boldsymbol{v}$: ${}^{B}\boldsymbol{q}_{\boldsymbol{v}} = [0 \quad {}^{B}\boldsymbol{v}]$
2. Rotate by quaternion ${}^{A}\boldsymbol{q}_{B}$:

$${}^{A}\boldsymbol{q}_{\boldsymbol{v}} = {}^{A}\boldsymbol{q}_{B}\ {}^{B}\boldsymbol{q}_{\boldsymbol{v}}\ {}^{A}\boldsymbol{q}_{B}^{*} = [0 \quad {}^{A}\boldsymbol{v}]$$

$${}^{B}\boldsymbol{q}_{\boldsymbol{v}} = {}^{B}\boldsymbol{q}_{A}\ {}^{A}\boldsymbol{q}_{\boldsymbol{v}}\ {}^{B}\boldsymbol{q}_{A}^{*} = {}^{A}\boldsymbol{q}_{B}^{*}\ {}^{A}\boldsymbol{q}_{\boldsymbol{v}}\ {}^{A}\boldsymbol{q}_{B} = [0 \quad {}^{B}\boldsymbol{v}]$$

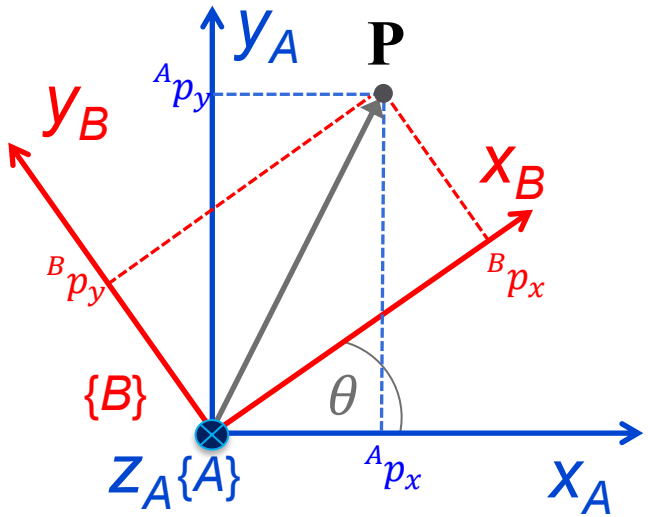
qauternion_vector_rotation.mlx

Describing a vector before and after rotation

From last lecture:

$$\boldsymbol{q}^{-1} = \boldsymbol{q}^{*} = [q_0 \quad -q_1 \quad -q_2 \quad -q_3]$$

$$\left({}^{B}\boldsymbol{q}_{A}\right)^{-1} = {}^{A}\boldsymbol{q}_{B}$$



$${}^{A}\boldsymbol{p} = \begin{bmatrix} {}^{A}p_x \\ {}^{A}p_y \\ {}^{A}p_z \end{bmatrix} \qquad {}^{B}\boldsymbol{p} = \begin{bmatrix} {}^{B}p_x \\ {}^{B}p_y \\ {}^{B}p_z \end{bmatrix}$$

With quaternions:

1. make quaternion from vector $\boldsymbol{v}$: ${}^{B}\boldsymbol{q}_{\boldsymbol{v}} = [0 \quad {}^{B}\boldsymbol{v}]$
2. Rotate by quaternion ${}^{A}\boldsymbol{q}_{B}$:

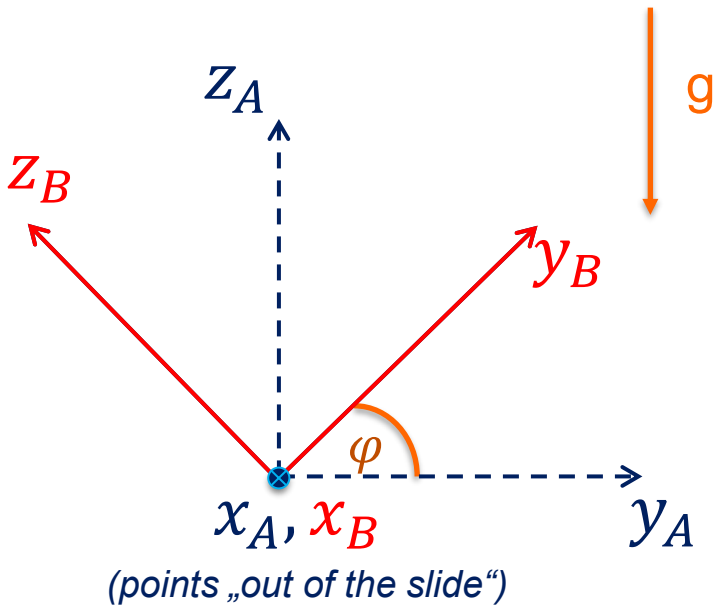
   Alternatively: ${}^{A}\boldsymbol{q}_{B} \rightarrow {}^{A}\boldsymbol{R}_{B}$

$$\boxed{{}^{A}\boldsymbol{v} = {}^{A}\boldsymbol{R}_{B}\ {}^{B}\boldsymbol{v}}$$

$$\boxed{{}^{B}\boldsymbol{v} = {}^{A}\boldsymbol{R}_{B}^{-1}\ {}^{A}\boldsymbol{v} = {}^{B}\boldsymbol{R}_{A}\ {}^{A}\boldsymbol{v}}$$

3D acceleration data

- How does the reading change when the sensor is rotated by an **arbitrary angle** φ about the *x*-axis?



$$^{B}\boldsymbol{a}_{\mathrm{m}} = [?\quad ?\quad ?] = \left[0 \quad \frac{\sqrt{2}}{2}g \quad \frac{\sqrt{2}}{2}g\right] \; ^{B}\boldsymbol{R}_{A}$$

$$^{A}\boldsymbol{g} = [0 \quad 0 \quad g]^{T} = {}^{A}\boldsymbol{a}_{\mathrm{m}} \quad \Rightarrow \quad ^{B}\boldsymbol{g} = \boxed{^{A}\boldsymbol{R}_{B}^{-1}}\,^{A}\boldsymbol{g}$$

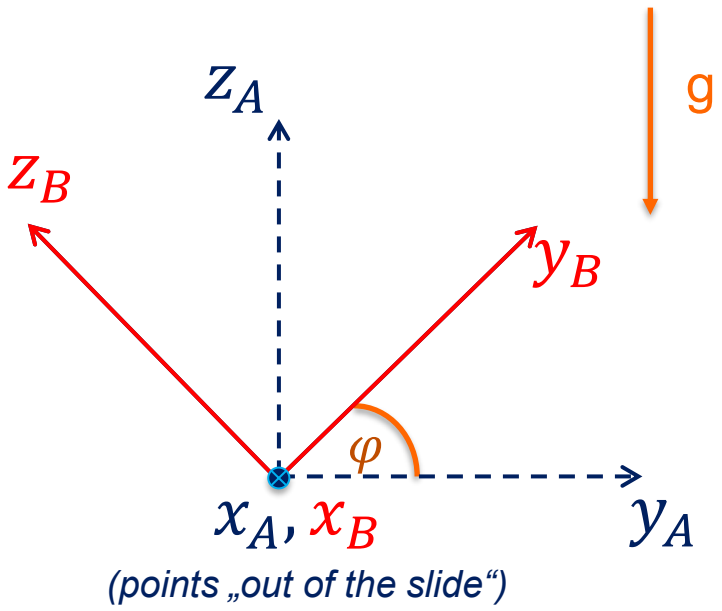
$$^{A}\boldsymbol{R}_{B} = \boldsymbol{R}_{x}(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix} \qquad ^{A}\boldsymbol{R}_{B}^{-1} = {}^{A}\boldsymbol{R}_{B}^{T} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{bmatrix}$$

Here: $\varphi = \frac{\pi}{4}$

$$^{B}\boldsymbol{g} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\frac{\pi}{4} & \sin\frac{\pi}{4} \\ 0 & -\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \mathrm{g} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{\sqrt{2}}{2}\mathrm{g} \\ \frac{\sqrt{2}}{2}\mathrm{g} \end{bmatrix} = \begin{bmatrix} 0 \\ 0.7071\mathrm{g} \\ 0.7071\mathrm{g} \end{bmatrix}$$

3D acceleration data

- How does the reading change when the sensor is rotated by an **arbitrary angle** φ about the *x*-axis?



$$^{B}\boldsymbol{a}_{\mathrm{m}} = [?\quad ?\quad ?] = \left[0 \quad \frac{\sqrt{2}}{2}g \quad \frac{\sqrt{2}}{2}g\right]$$

In any case, under stationary conditions:

$$\boxed{|\boldsymbol{a}_{\mathrm{m}}| = \sqrt{\boldsymbol{a}_{\mathrm{m}_x}^2 + \boldsymbol{a}_{\mathrm{m}_y}^2 + \boldsymbol{a}_{\mathrm{m}_z}^2} = g}$$

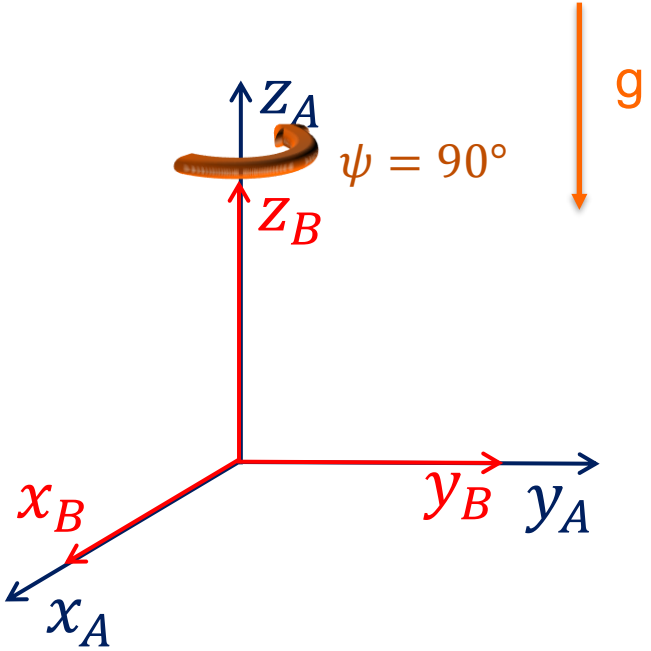
e.g.:

$$\sqrt{0 + \left(\frac{\sqrt{2}}{2}g\right)^2 + \left(\frac{\sqrt{2}}{2}g\right)^2} = \sqrt{0 + \frac{2}{4}\mathrm{g}^2 + \frac{2}{4}g^2} = g$$

3D acceleration data

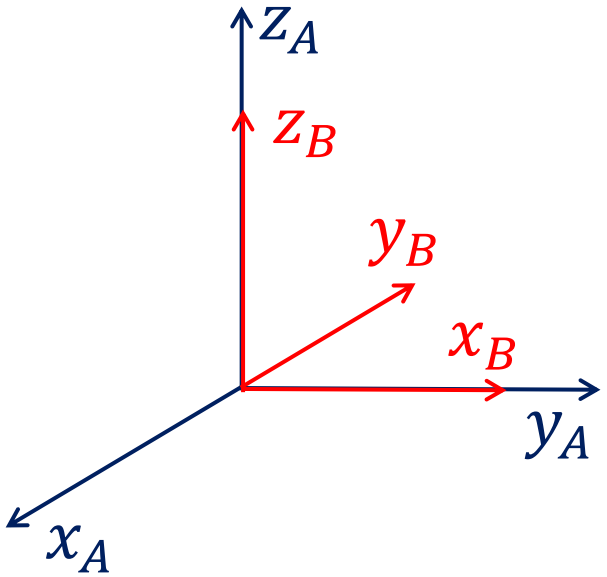
What happens if sensor is rotated to an axis parallel to the gravity vector?

Before rotation



${}^{B}\boldsymbol{a}_{\mathrm{m}} = [0 \quad 0 \quad g]$

After rotation



${}^{B}\boldsymbol{a}_{\mathrm{m}} = [? \quad ? \quad ?]$

No change in measurement as rotation axis is parallel to vector [0 0 g]!

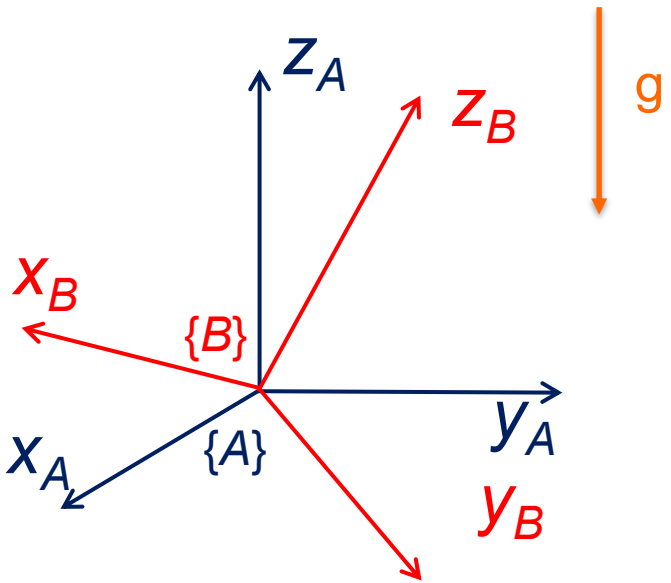
$$ {}^{A}\boldsymbol{R}_{B} = \boldsymbol{R}_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{A}\boldsymbol{R}_{B}^{-1} = \boldsymbol{R}_z{}^{T}(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{B}\boldsymbol{g} = {}^{A}\boldsymbol{R}_{B}^{-1}\,{}^{A}\boldsymbol{g} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} $$

$$ = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = {}^{A}\boldsymbol{g} $$

Determining the roll and pitch angle based on acceleration measurements



$$^{B}\boldsymbol{g} = {}^{B}\boldsymbol{a}_{\mathrm{m}} = {}^{A}\boldsymbol{R}_{B}^{-1}\,{}^{A}\boldsymbol{a}_{\mathrm{m}} = {}^{A}\boldsymbol{R}_{B}^{-1}\,{}^{A}\boldsymbol{g} = {}^{A}\boldsymbol{R}_{B}^{-1}\begin{bmatrix}0\\0\\\mathrm{g}\end{bmatrix}$$

Goal:
Determine roll and pitch based on accelerometer reading

Approach:
1. Find mathematical expression for ${}^{A}\boldsymbol{R}_{B}^{-1}$ as combinations of R_x, R_y and R_z
2. Solve for roll and pitch angles

Determining the roll and pitch angle based on acceleration measurements

$$\mathrm{R}_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix} \quad \mathrm{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \mathrm{R}_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\overbrace{\left(\mathrm{R}_z(\psi)\mathrm{R}_y(\theta)\mathrm{R}_x(\varphi)\right)^T}^{{}^{A}\boldsymbol{R}_B^{-1}} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(normalized measurment values via $\frac{\boldsymbol{a}_m}{|\boldsymbol{a}_m|}$)

$$= \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta\sin\varphi - \cos\varphi\sin\psi & \sin\psi\sin\varphi + \cos\psi\cos\varphi\sin\theta \\ \cos\theta\sin\psi & \cos\theta\cos\varphi & \cos\varphi\sin\psi\sin\theta - \cos\psi\sin\varphi \\ -\sin\theta & \cos\theta\sin\varphi & \cos\theta\cos\varphi \end{bmatrix}^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi\cos\theta & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\varphi - \cos\varphi\sin\psi & \cos\theta\cos\varphi & \cos\theta\sin\varphi \\ \sin\psi\sin\varphi + \cos\psi\cos\varphi\sin\theta & \cos\varphi\sin\psi\sin\theta - \cos\psi\sin\varphi & \cos\theta\cos\varphi \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta\sin\varphi \\ \cos\theta\cos\varphi \end{bmatrix}$$

Determining the roll and pitch angle based on acceleration measurements

$$\begin{bmatrix} -\sin\theta \\ \cos\theta\sin\varphi \\ \cos\theta\cos\varphi \end{bmatrix} = {}^{B}\boldsymbol{a}_{norm} = \boldsymbol{a}_{norm} = \frac{\boldsymbol{a}_m}{|\boldsymbol{a}_m|} = \frac{1}{\sqrt{a_{m_x}^2 + a_{m_y}^2 + a_{m_z}^2}} \begin{bmatrix} a_{m_x} \\ a_{m_y} \\ a_{m_z} \end{bmatrix}$$

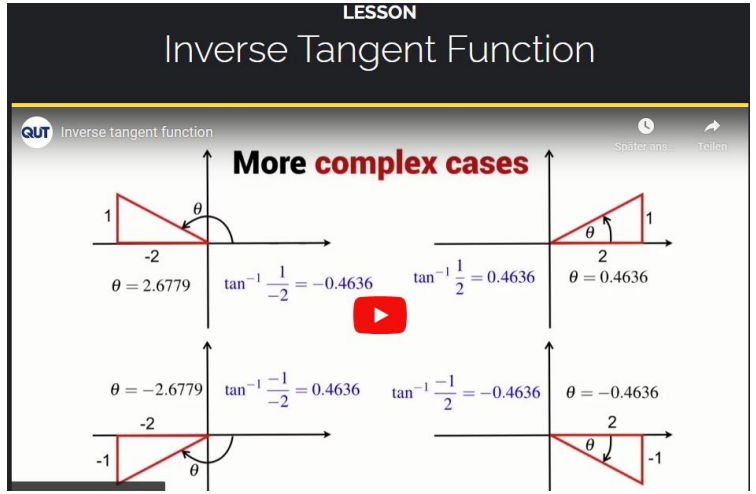
$$\boxed{\tan\varphi = \frac{\boldsymbol{a}_{norm_y}}{\boldsymbol{a}_{norm_z}}} = \frac{\cos\theta\sin\varphi}{\cos\theta\cos\varphi} = \frac{\sin\varphi}{\cos\varphi}$$

$$\boxed{\tan\theta = \frac{-\boldsymbol{a}_{norm_x}}{\sqrt{\boldsymbol{a}_{norm_y}^2 + \boldsymbol{a}_{norm_z}^2}}} = \frac{\sin\theta}{\sqrt{\cos^2\theta\sin^2\varphi + \cos^2\theta\cos^2\varphi}} = \frac{\sin\theta}{\sqrt{\cos^2\theta\,\underbrace{(\sin^2\varphi + \cos^2\varphi)}_{1}}} = \frac{\sin\theta}{\cos\theta}$$

Determining the roll and pitch angle based on acceleration measurements

$$\tan\varphi = \frac{\boldsymbol{a}_{norm_y}}{\boldsymbol{a}_{norm_z}} \quad \Rightarrow \quad \varphi = \operatorname{atan2}\left(\boldsymbol{a}_{norm_y}, \boldsymbol{a}_{norm_z}\right)$$

$$\tan\theta = \frac{-\boldsymbol{a}_{norm_x}}{\sqrt{\boldsymbol{a}_{norm_y}^2 + \boldsymbol{a}_{norm_z}^2}} \quad \Rightarrow \quad \theta = \operatorname{atan2}\left(-\boldsymbol{a}_{norm_x}, \sqrt{\boldsymbol{a}_{norm_y}^2 + \boldsymbol{a}_{norm_z}^2}\right)$$



https://robotacademy.net.au/lesson/inverse-tangent-function/

3D Accelerometer

What does the measurement contain in case of „external“ acceleration?

$$\boldsymbol{a}_{\mathrm{m}} = [0 \quad 0 \quad g] + [a_x \quad a_y \quad a_z]$$

where a_x, a_y and a_z are the components of the external acceleration vector $\boldsymbol{a}_{ext}$

$$\cancel{{}^{B}\boldsymbol{g} = {}^{A}\boldsymbol{R}_{B}^{-1}\,{}^{A}\boldsymbol{a}_{\mathrm{m}} = {}^{A}\boldsymbol{R}_{B}^{-1}\,{}^{A}\boldsymbol{g}} \quad \text{as } {}^{A}\boldsymbol{a}_{\mathrm{m}} \neq {}^{A}\boldsymbol{g}$$

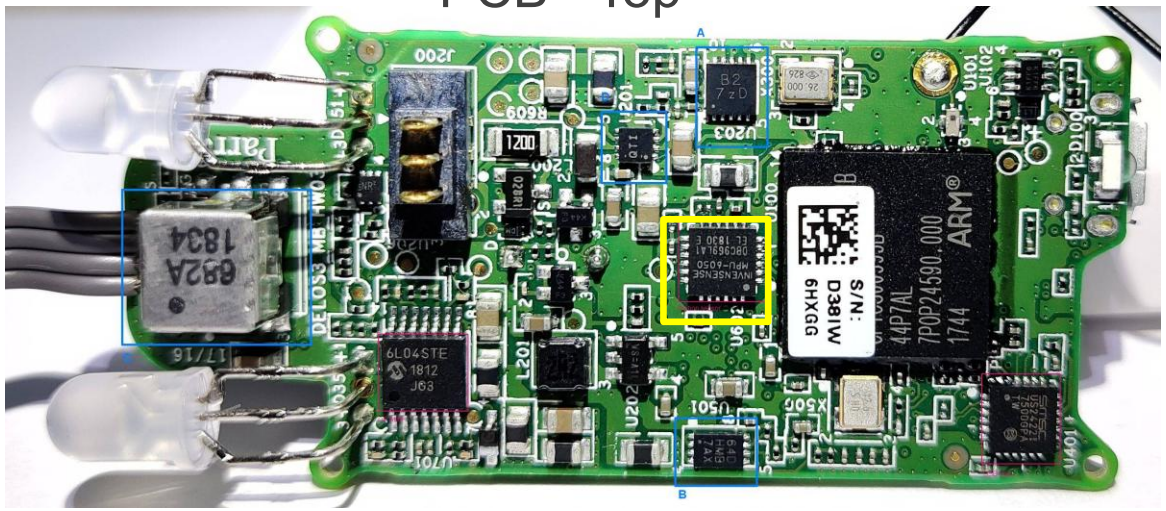
$$\cancel{|\boldsymbol{a}_{\mathrm{m}}| = \sqrt{\boldsymbol{a}_{\mathrm{m}x}^2 + \boldsymbol{a}_{\mathrm{m}y}^2 + \boldsymbol{a}_{\mathrm{m}z}^2} = g}$$

$$|\boldsymbol{a}_{\mathrm{m}}| = \sqrt{\boldsymbol{a}_{\mathrm{m}x}^2 + \boldsymbol{a}_{\mathrm{m}y}^2 + \boldsymbol{a}_{\mathrm{m}z}^2 + \boldsymbol{a}_{x}^2 + \boldsymbol{a}_{y}^2 + \boldsymbol{a}_{z}^2}$$

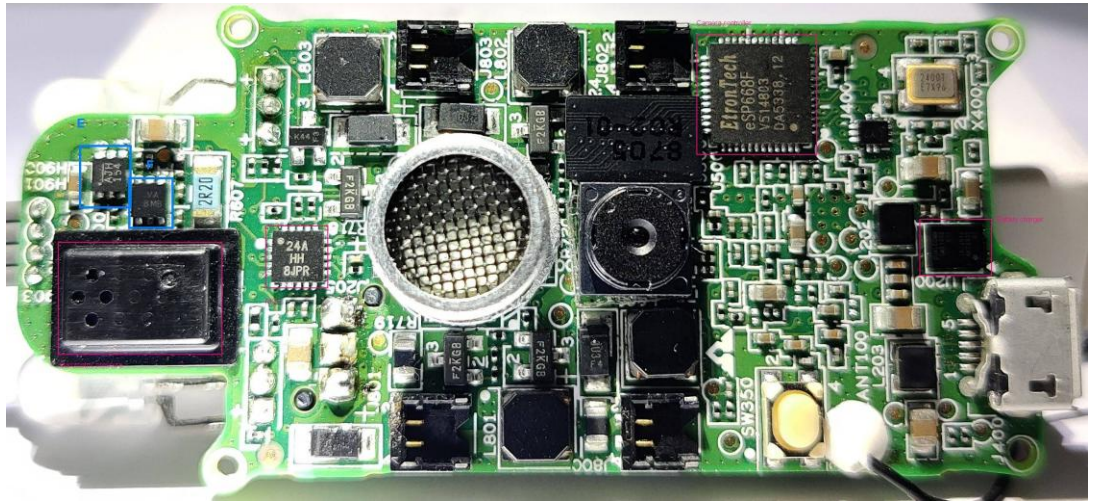
Under dynamic (=non static) conditions, accelerometer data cannot be used for reliable orientation estimation!

Exercise L4.1 – Parrot Mambo axes

PCB - Top



PCB - Bottom



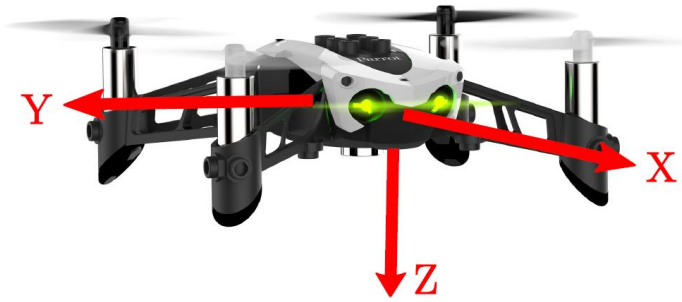
- Check the datasheet of the MPU-6050: how are the axes directions defined?

Exercise L4.2 – rotation angles

- Implement the equations for roll and pitch calculation.
  In order to account for the mambos coordinate system, **use these equations**:

$$\varphi = -\mathrm{atan2}\left(\boldsymbol{a}_{norm_y}, -\boldsymbol{a}_{norm_z}\right)$$

$$\theta = -\mathrm{atan2}\left(-\boldsymbol{a}_{norm_x}, \sqrt{\boldsymbol{a}_{norm_y}^2 + \boldsymbol{a}_{norm_z}^2}\right)$$



- download the mat-files containing sensor-data measured by the mambo drone from ILIAS

- Apply the functions to the data in the mat-files and generate a figure with two subplots: one for acceleration and one for roll and pitch. Scale the y-axes of the first subplot to [-10… 10]m/s^2 and the y-axes of the second subplot to [-100…100]°

- How was the drone rotated regarding *x*- (roll) and *y*-axis (pitch) during the measurements?